

# 1 Matplotlib Basics

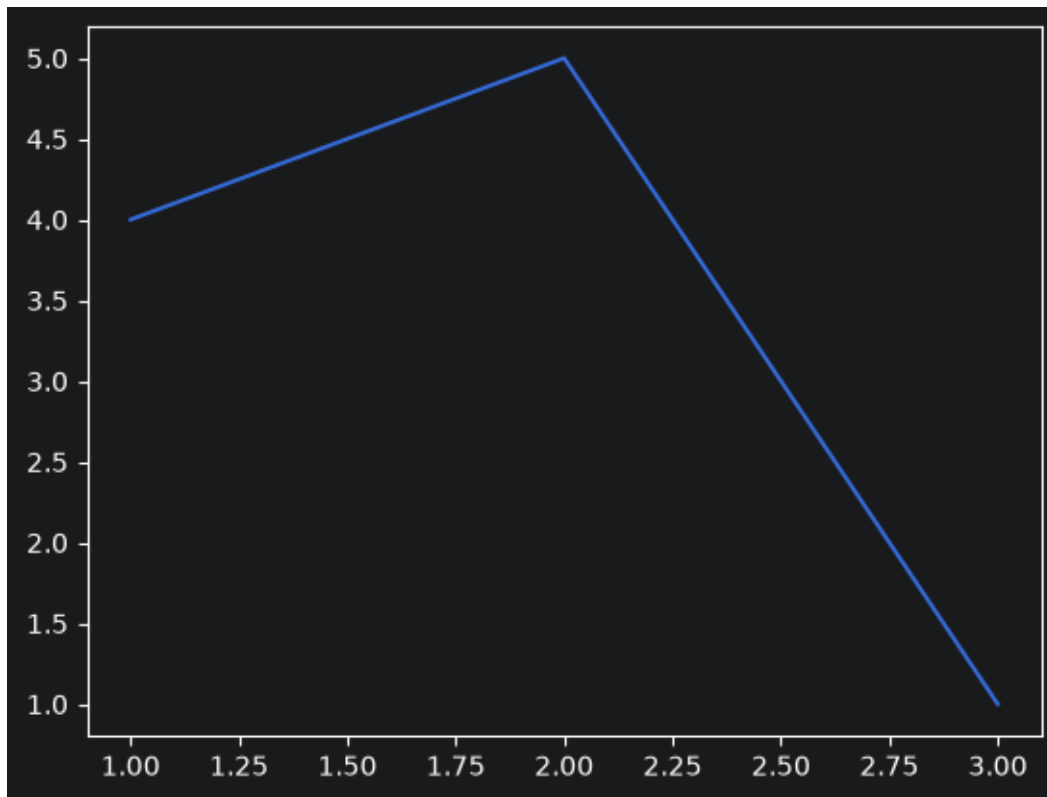
## 1.1 Our first Visualization

```
[1]: # Importing pyplot
# matplotlib is a visualization library, pyplot is a module that contains function
# for creating graphs

# from matplotlib import pyplot as plt
import matplotlib.pyplot as plt

# Plotting to our canvas
plt.plot([1, 2, 3], [4, 5, 1])

# Showing what we plotted
plt.show()
```



## 1.2 Adding Titles and Labels

```
[2]: import matplotlib.pyplot as plt

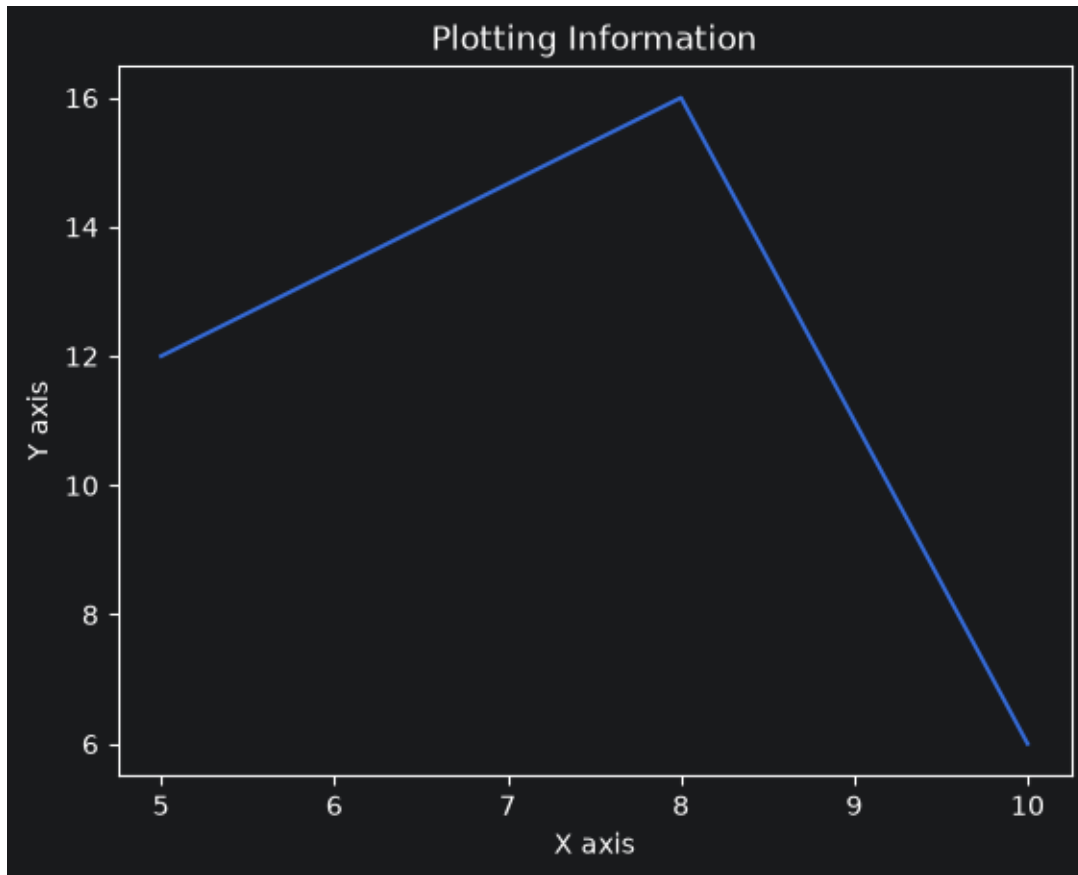
x = [5, 8, 10]
y = [12, 16, 6]
plt.plot(x, y)
```

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```
plt.title("Plotting Information")

# Putting label to both the axis
plt.ylabel("Y axis")
plt.xlabel("X axis")

plt.show()
```



### 1.3 Adding Styling

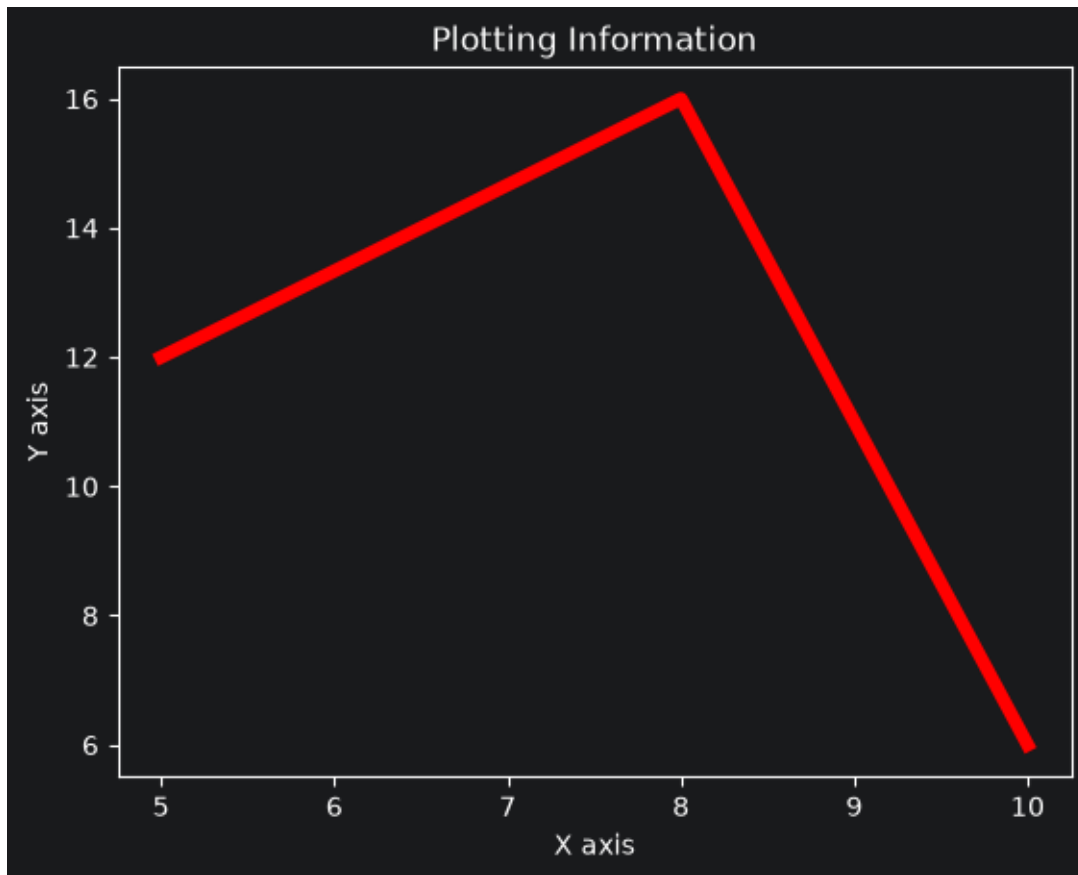
```
[3]: import matplotlib.pyplot as plt
import matplotlib.style

x = [5, 8, 10]
y = [12, 16, 6]
plt.plot(x, y, color="red", linewidth=5)
plt.title("Plotting Information")

# Putting label to both the axis
plt.ylabel("Y axis")
plt.xlabel("X axis")
```

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```
plt.show()
```



## 1.4 Saving it to File

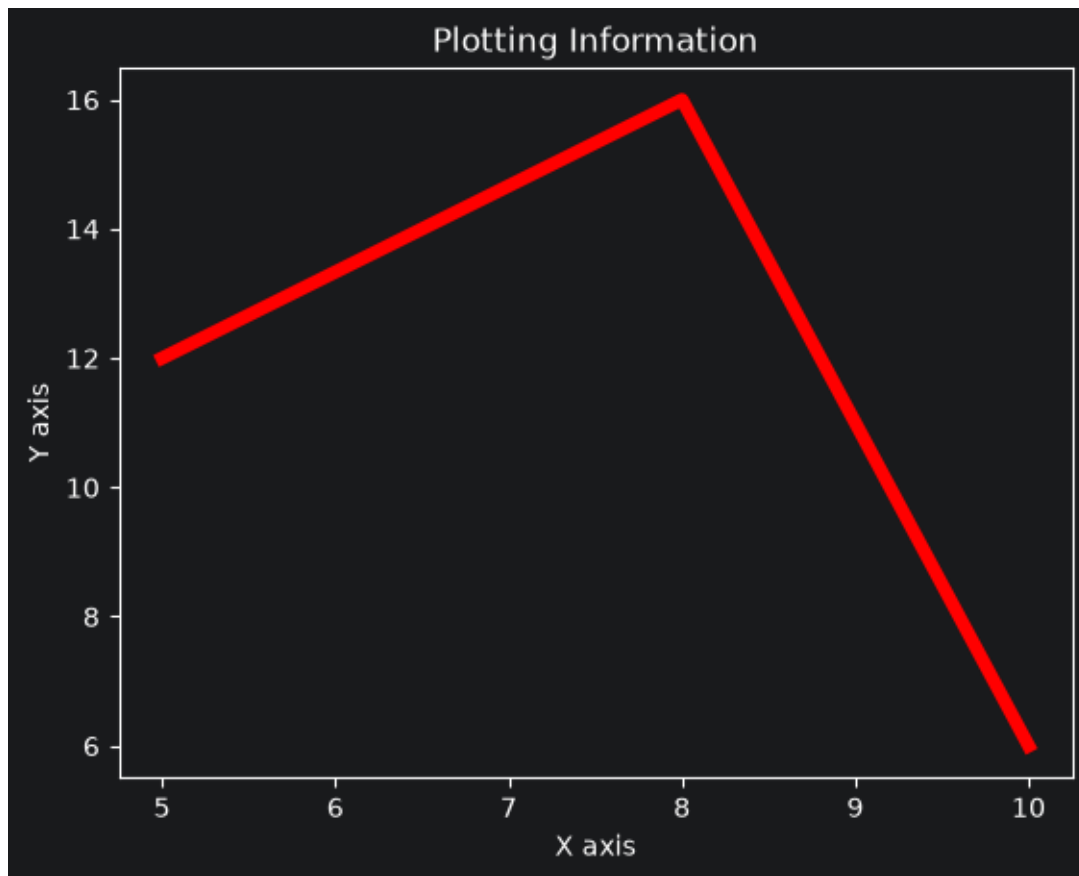
```
[4]: import matplotlib.pyplot as plt
import matplotlib.style

x = [5, 8, 10]
y = [12, 16, 6]
plt.plot(x, y, color="red", linewidth=5)
plt.title("Plotting Information")

# Putting label to both the axis
plt.ylabel("Y axis")
plt.xlabel("X axis")

plt.savefig("MyPlot.png", format="png")
```

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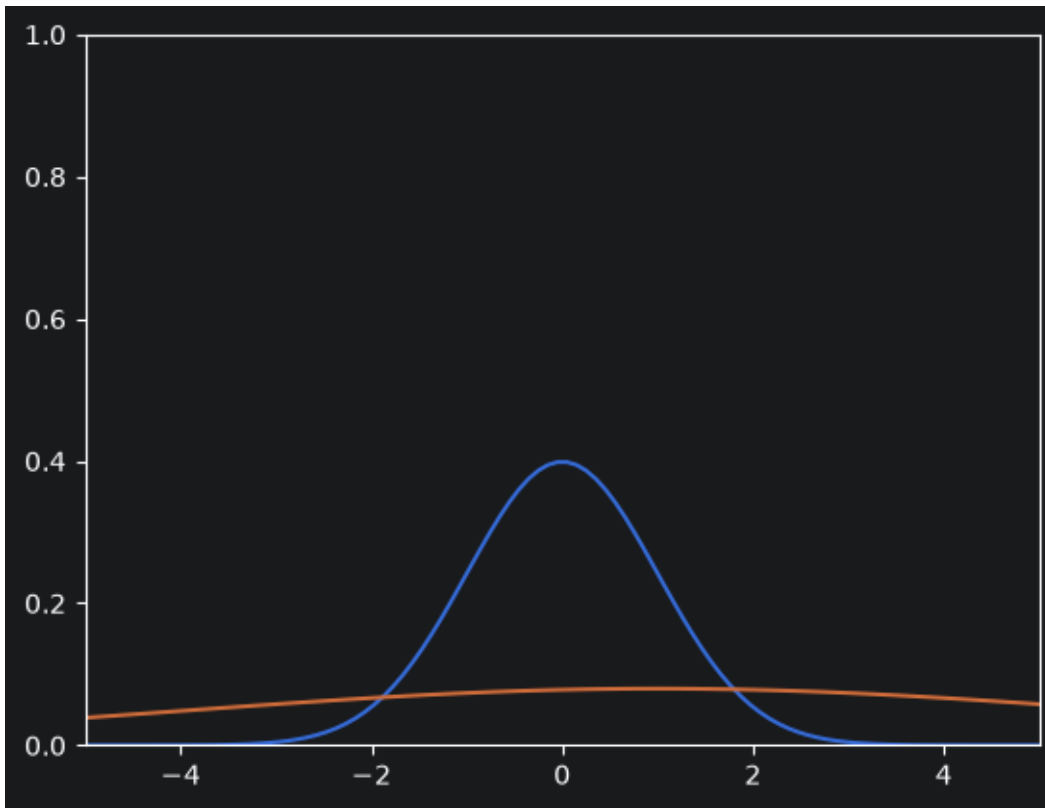


```
[5]: import numpy as np
from scipy.stats import norm

axes = plt.axes()
axes.set_xlim([-5, 5])
axes.set_ylim([0, 1.0])
# axes.set_xticks([-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5])
# axes.set_yticks([0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0])

x = np.linspace(-5, 5, 100)
plt.plot(x, norm.pdf(x))
plt.plot(x, norm.pdf(x, 1.0, 5.0)) # mean = 1.0 and standard deviation = 0.5
plt.show()
```

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